



Figure 5-6: Suggested project folder hierarchy

A *path* (also called a *pathname*) is a string that uniquely identifies the location of a file or folder on a computer. To load a graphic or sound file into your program, you must specify the path to the file. There are two types of paths: relative and absolute.

A *relative path* is a relative to the current folder, often called the *current working directory*. When you run a program using an IDE such as IDLE or PyCharm, it sets the current folder to the one containing your main Python program so you can use relative paths with ease. In this book, I will assume you're using an IDE and will represent all paths as relative paths.

The relative path for a graphic file (for example, *ball.png*) in the same folder as your main Python file would be just the filename as a string (for example, 'ball.png'). Using the suggested project structure, the relative path would be 'images/ball.png'.

This says that inside the project folder will be another folder named *images*, and inside that folder is a file named *ball.png*. In path strings, folder names are separated by the slash character.

However, if you expect to run your program from the command line, then you need to construct absolute paths for all files. An *absolute path* is one that starts from the root of the filesystem and includes the full hierarchy of folders to your file. To build an absolute path to any file, you can use code like this, which builds an absolute path string to the *ball.png* file in the *images* folder inside the project folder:

```
from pathlib import Path

# Place this in section #2, defining a constant
BASE_PATH = Path(__file__).resolve().parent
```